

Total suspended solids as a simplified metric for predicting optimized dissolved oxygen in wastewater treatment: Comparing process-based and artificial neural network models

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ABSTRACT

Freshwater scarcity is an escalating global issue, thus requiring effective solutions such as water reuse. However, achieving water reuse depends on efficient wastewater treatment, which is highly energy intensive. In a typical wastewater treatment plant (WWTP), aeration accounts for the majority of the energy consumption, which is generally not optimized according to the influent flow and water quality characteristics. This study investigates the potential of artificial neural networks (ANNs) to predict optimized dissolved oxygen (DO) levels in aeration tank for reducing energy consumption in a sewage treatment plant (STP). Using experimentally collected historical data from the already-optimized Chikkabanavara STP in Bangalore, two ANN models were developed and compared, one using mass flow rate of total suspended solids (TSS) as the sole input parameter, and another using a combination of mass flow rates of TSS, biochemical oxygen demand (BOD), and chemical oxygen demand (COD) as inputs. Both models exhibited similar predictive performance for DO with robust accuracy (Robust $R^2 = 0.99$, $R^2 = 0.87$). The predicted DO was proved to effectively treat the wastewater, and it was validated using a digital twin of STP by process-based BioWin simulations. The results highlight that using TSS as a proxy for COD and BOD not only simplifies monitoring but also enhances process control, ensuring efficient operation and compliance with discharge standards.

1. Introduction

The global scarcity of freshwater is intensifying [1], particularly in regions with limited water resources, driving the need for effective water reuse strategies. In India, and other water-stressed nations, wastewater treatment and recycling are essential to preserve freshwater reserves, safeguarding public health, and protecting the environment. In places like Kolar, approximately 440 million litres per day (MLD) of water is recycled to mitigate the effects of recurrent droughts, exemplifying a growing trend of water reuse in regions facing acute water scarcity [2]. Thus, energy-efficient wastewater treatment is the need of the hour.

However, municipal wastewater treatment plants (WWTPs) are energy-intensive facilities, consuming substantial energy to power aeration systems, pumps, and sludge treatment processes. According to

the United States Environmental Protection Agency (EPA), energy usage in WWTPs typically constitutes 25–40 % of their total operational costs, making energy optimization a crucial focus for plant management [3]. This high energy demand not only drives up operational expenses but also contributes significantly to carbon emissions, thus exacerbating climate change. A study surveying energy use across WWTPs found an average consumption of 0.7 ± 0.6 kWh/m³, with certain countries including France, Japan, Germany, and, the USA showing particularly high energy demands [4–9].

In WWTPs, aerations systems consume the 45–75 % of the total plant energy [10]. Most WWTPs operate with a fixed dissolved oxygen (DO) level, regardless of variations in influent flow and water quality characteristics. This means that whether the influent is of high or low strength, the aeration DO level remains constant [11]. Hence DO needs to be optimized accordingly for energy efficient treatment systems. In

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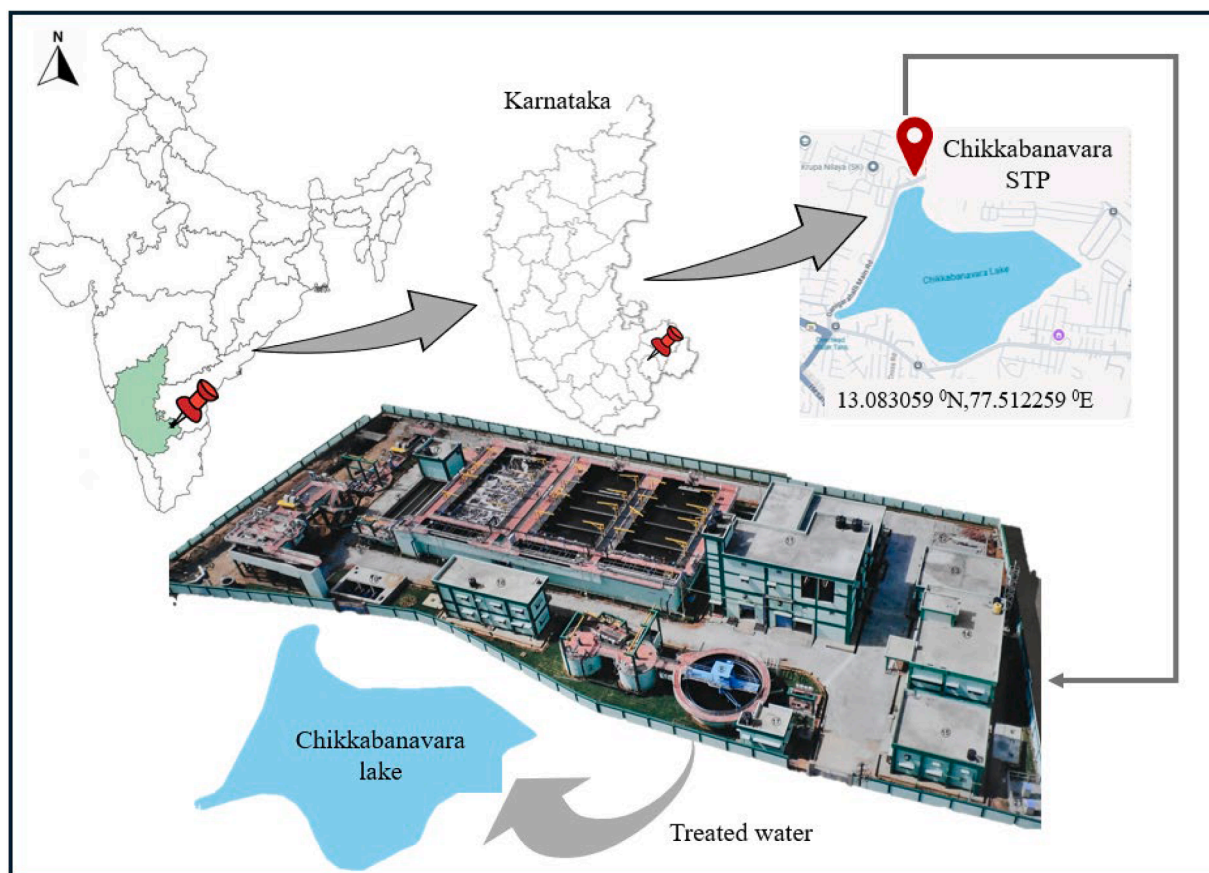


Fig. 1a. Chikkabanavara STP: location.

recent years, unlike traditional models, artificial intelligence and machine learning (AI ML) techniques have been able to detect complex patterns and adapt to varying wastewater characteristics, presenting new possibilities for DO optimization and predictive maintenance. Researchers [12–14] have explored several AI–ML methods for optimizing DO in WWTPs for energy effective treatments. However, training these artificial neural network (ANN) models requires a substantial amount of data, which can only be obtained through on-field experimentation [15]. This process is both time-consuming and chemical-intensive, as it necessitates extensive real-world data collection and analysis to ensure the models are accurate and reliable. Though several ANN models exist, real-world on-field implementation is rarely observed [16–23]. This is primarily because most ANN models require multiple input parameters, such as Biochemical Oxygen Demand (BOD), Chemical Oxygen Demand (COD), Total Phosphorus (TP), and Total Nitrogen (TN), which are difficult to measure in real-time. These parameters often require lab analysis, making them impractical for continuous monitoring WWTPs. As a result, using such models for real-time DO optimization can be challenging, as obtaining all necessary inputs delays decision-making. Additionally, the time required to process these inputs and adjust operational settings can slow down the startup and efficiency of WWTPs, making real-time optimization less feasible in practical applications.

Thus, this study aims to simplify the process of training ANN models for predicting optimized DO of aeration tank, in sewage treatment plants (STPs), using minimal and easy to measure input parameters. The goal is to determine whether the ANN models can accurately predict the optimum DO required in an aeration tank to treat the wastewater, with just a single input parameter, ensuring that treated water meets discharge standards. This will be verified using BioWin, a process-based software that simulates biochemical processes and offers scenario-based projections water treatment efficiency. This study will compare the

performance of ANN models, emphasizing the flexibility, efficiency, and adaptability of ANN-based approaches for real-time energy optimization.

2. Methodology

2.1. Study area

Water samples and energy data were collected from the Chikkabanavara sewage treatment plant (5 million liters per day (MLD)), located near Chikkabanavara Lake in the Arkavathi division of Bangalore city, India (Fig 1a). The Chikkabanavara STP was commissioned in 2018 and was contracted and maintained by M/s Suez. The plant is currently receiving on average 4.5 MLD sewage. The treatment plant consists of three Sequential Batch Reactor (SBR) basins, two of which operate in tandem on a 3-hour cycle under average flow conditions. During peak flow periods, characterized by a peak factor of 1.05, the cycle time is extended to 4 h. The plant is designed to store and treat excess sewage within the SBR chambers. The design values for COD and BOD are 800 mg/L and 350 mg/L, respectively. The Mixed Liquor Suspended Solids (MLSS) concentration in the SBR chambers is maintained at 3500 mg/L. The aeration control strategy in this STP was optimized prior, as per flow and influent water quality, using BioWin simulations. Aeration is provided through diffused aerators, which are manually cleaned and backwashed every six months. The SBR basins are cleaned every two months to maintain operational efficiency. The plant operates with a Return Activated Sludge (RAS) rate of 40 %, with RAS and Waste Activated Sludge (WAS) volumes of 95 m³/hr and 49 m³/hr, respectively. In the tertiary treatment stage, chlorine is dosed to achieve a final concentration of 0.2–0.3 mg/L. The effluent is discharged into the adjacent Chikkabanavara Lake. Sludge generated at the plant is utilized

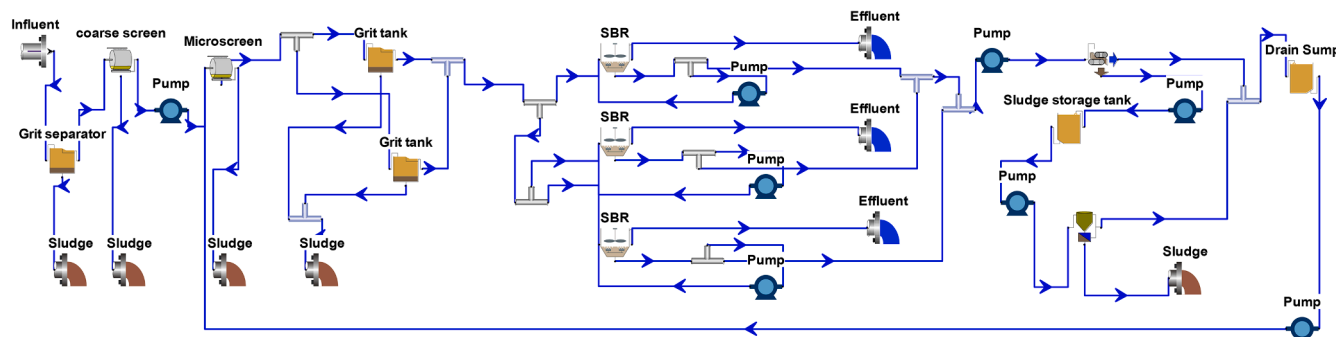


Fig. 1b. Chikkabanavara STP digital twin on BioWin: Process flow diagram.

Table 1

Summary statistics of key parameters.

	Inflow (MLD)	DO (mg/L)	pH	Temperature (°C)	COD (mg/L)	BOD (mg/L)	TSS (mg/L)	NH4-N (mg/L)	Total Phosphorous (mg/L)
Min	1.48	0.96	6.57	25.80	580.00	290.00	200.00	21.80	2.90
Max	5.11	5.60	7.26	27.9	706.00	320.00	248.00	27.2	4.50
Avg	3.63	3.78	6.93	27.06	634.79	304.44	222.89	25.02	3.50
Std dev	0.84	0.61	0.15	0.36	36.21	8.77	14.42	1.46	0.43

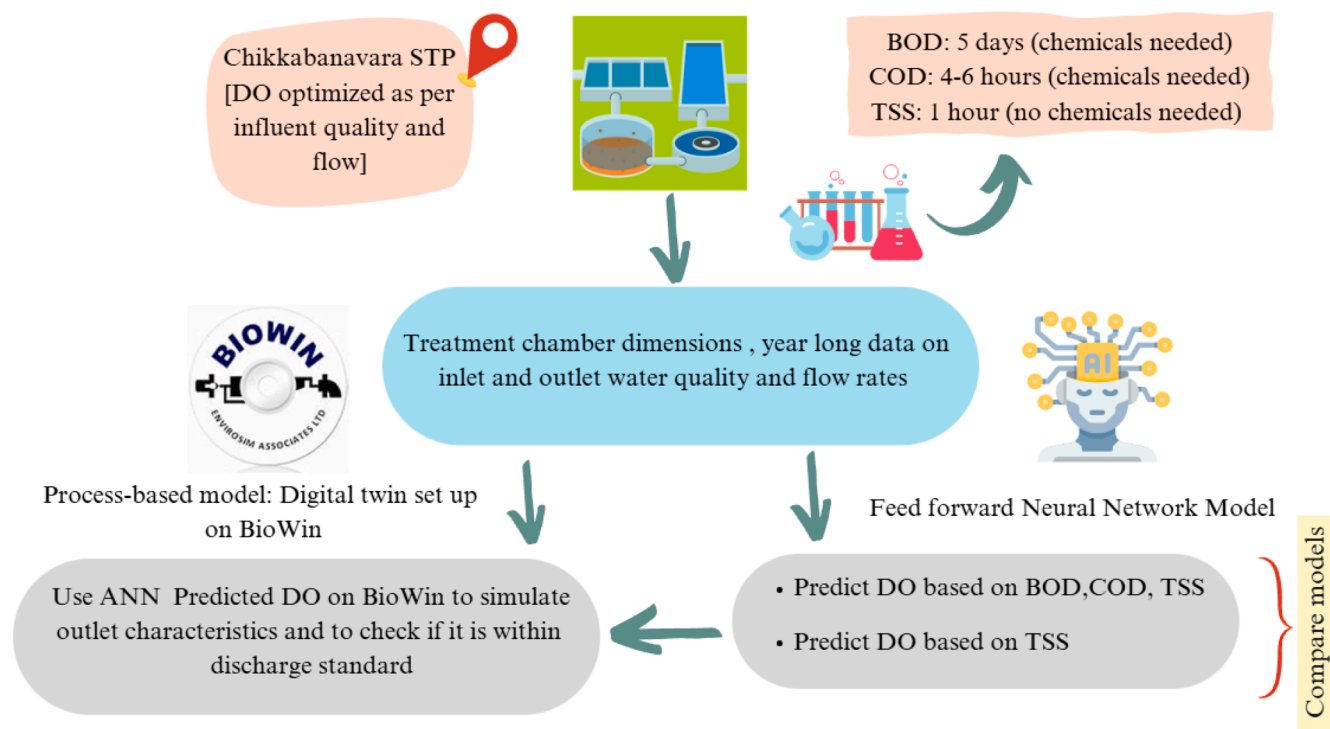


Fig. 2. Study framework.

as manure. A process flow diagram of Chikkabanavara STP is shown in Fig 1b.

2.2. Data collection

Parameters including BOD, COD, TSS, nutrient concentrations (such as nitrogen and phosphorus), were assessed via grab sampling conducted daily over a duration of one year. In addition, the flow to the STP was continuously monitored and recorded to provide insights into treatment dynamics. Influent and effluent water quality assessment was carried out following the protocols established by the American Public

Health Association [24]. The summary statistics of inlet wastewater characteristics and flow rates are reported in Table 1.

2.3. Modelling framework: black box ANN and process-based BioWin

ANNs and mechanistic models like BioWin offer distinct approaches to wastewater treatment modeling. ANNs, a powerful predictive tool, is capable of capturing complex, nonlinear relationships within datasets [25]. However, they function as black-box models, lacking interpretability regarding the underlying biochemical and physicochemical processes and hence generalization becomes difficult [26]. BioWin, on

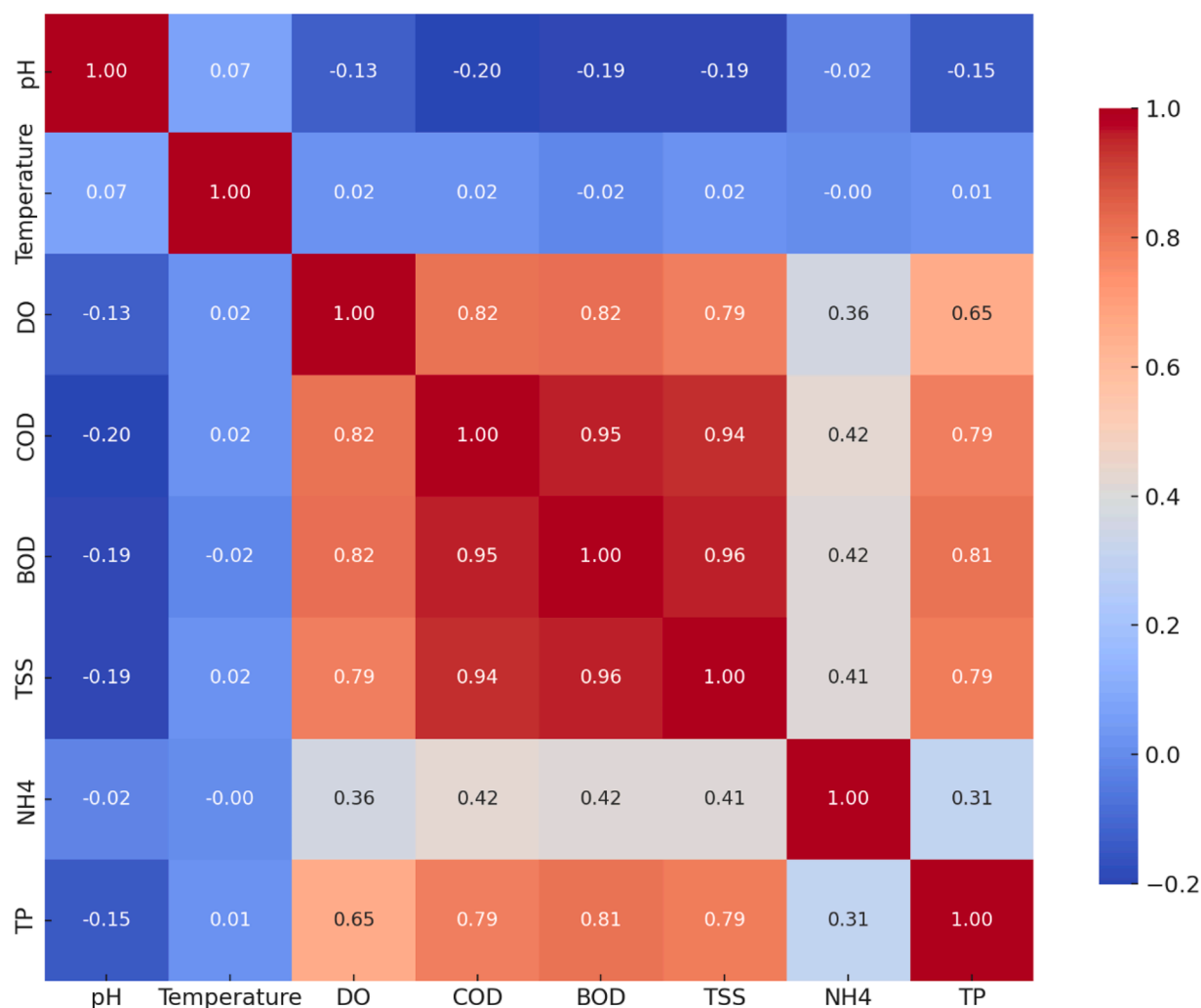


Fig. 3. 3 Spearman's Rho correlation coefficients for key parameters.

the other hand, is a mechanistic model that incorporates basic principles of wastewater treatment, simulating microbial kinetics, oxygen dynamics, and substrate usage based on derived process equations [27].

An ANN model was first developed using data collected from the Chikkabanavara STP, where DO levels had previously been optimized through BioWin software simulations. The model was trained to predict the optimal DO required in the aeration tank. To validate the practical relevance of the ANN predictions, the predicted DO values were input into BioWin simulations to evaluate their impact on effluent quality and overall process performance. This methodology helped to assess whether the ANN-predicted DO levels was sufficient to support key biological processes such as organic matter degradation, nitrification, and denitrification using a process-based model. Additionally, BioWin served as a benchmark to determine whether the effluent quality met regulatory discharge standards. Fig 2 graphically depicts the study framework. In the following sections the details of both ANN and BioWin model development is discussed.

2.3.1. ANN model: black box

To enhance energy efficiency in the aeration chambers of wastewater treatment facilities, this study used the to predict the optimized DO levels based on inflow rates, and various water quality parameters.

- Data Generation:** The data for the ANN model was collected over a year from Chikkabanavara STP, as mentioned in section 2.2. The Chikkabanavara STP was selected for model development as its DO levels had been previously optimized using BioWin simulations based on inflow characteristics. This ensured that the training data reflected optimal operational conditions, enabling the ANN model to learn and predict optimal DO levels in the aeration tank of STP for energy efficient treatment performance.
- Preprocessing:** The data underwent comprehensive preprocessing to ensure effective model development. Missing values were addressed through interpolation, where the missing data points were imputed using the average of the data points immediately preceding and following the gaps. This method helped maintain continuity and integrity in the dataset, ensuring reliable inputs for subsequent analysis.
- Choosing model parameters:** To select the most suitable input parameters for predicting DO levels, a correlation analysis was conducted, the results are shown in Fig. 3. The results revealed that COD (0.82), BOD (0.82), and TSS (0.79) exhibited strong correlations with DO, indicating their significant influence on oxygen depletion in water bodies. In contrast, parameters such as pH (−0.13), and NH₄ (0.36) showed weaker correlations, suggesting a lesser direct impact on DO levels. Based on these findings, COD, BOD, and TSS were

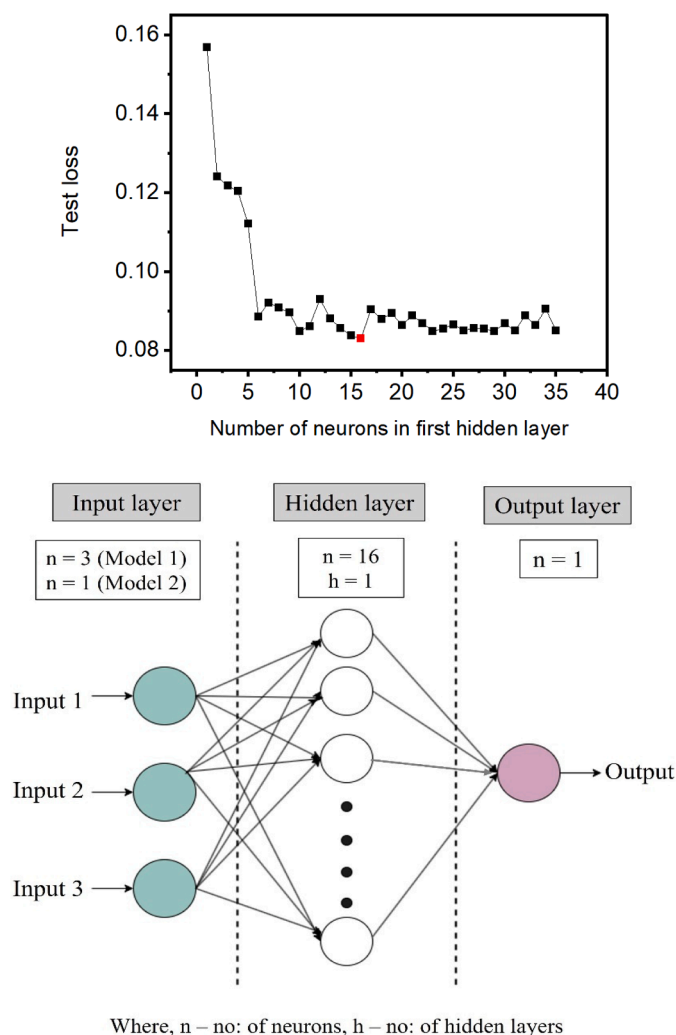


Fig. 4. a) Optimal number of neurons based on test loss. b) ANN model architecture.

Table 2

Adjusted parameters for the validated model.

Component	Significance of the component	Default	Value
F_{us} - Unbiodegradable soluble [gCOD/g of total COD]	Control the BOD to COD ratio. Alterations can help adjust the BOD based on input COD by controlling the unbiodegradable fraction.	0.05	0.03
F_{nus} - Soluble unbiodegradable TKN [g N/g TKN]	Control the TKN in effluent. Altering the component changes the effluent TKN to control Total nitrogen.	0.02	0.08
BOD calculation rate constant for X_{sc} degradation [1/d]	Control the BOD to COD ratio. Alterations can help adjust the BOD based on input COD	0.5	0.48
BOD calculation rate constant for X_{sc} degradation [1/d]	Control the BOD to COD ratio. Alterations can help adjust the BOD based on input COD	0.5	0.48

chosen as input parameters for the ANN model to ensure accurate DO predictions.

d. Model Development and Feature Engineering: The study used a feedforward backpropagation neural network to optimize DO in the aeration chambers of an STP. The input parameters for the model are:

- $[BOD\ inlet - BOD\ outlet] * flow\ rate$: Calculates the total mass of BOD removed from the wastewater per day (kg/day).
- $[COD\ inlet - COD\ outlet] * flow\ rate$: Represents the total mass of COD eliminated during treatment (kg/day).
- $[TSS\ inlet - TSS\ outlet] * flow\ rate$: Determines total amount of suspended solids removed daily (kg/day).
- $DO * flow\ rate$: Specifies the mass loading rate of dissolved oxygen.

These transformed values were normalized to scale all variables between 0 and 1, using min-max scaler. Normalization was particularly important because parameters such as BOD and COD typically exhibit significantly higher magnitudes compared to others like flow rate and DO. Without normalization, these high-magnitude variables could dominate the modelling process, leading to disproportionate weightage and biased predictions [28].

e. Data Partitioning: The data was split into 70 %, 15 %, and 15 % for training, validation, and testing, respectively. This split was selected after experimenting with various combinations including 60–20–20, 70–20–10, 75–15–10, 80–10–10, and 85–5–10. Among these, the 70–15–15 configuration consistently yielded better performance and generalization, making it the optimal choice for model development. A summary of the results from other split combinations is provided in the supplementary file (Table S1). This approach ensured that the model was trained on a large portion of the data, validated on a separate subset for fine-tuning, and tested on an entirely different set to assess its generalization ability.

f. Model defining: Two ANN models were designed to predict optimum DO concentrations in aeration, tailored to address varying operational and expertise requirements:

Model 1. A model incorporating COD, BOD, TSS, and flow rates as input to predict optimum DO. This model is for scenarios where expert oversight and comprehensive data infrastructure are available.

Model 2. A user-friendly, simplified model utilizing only TSS and flow rates as input to predict optimum DO levels. The model has parameters that are cost-effective and straightforward to measure, requiring minimal technical expertise. This model is ideal for resource-constrained environments, field-based monitoring, or rapid assessments.

The ANN models designed for this study consists of a simple feed-forward architecture with one hidden layer. The input layer takes in one/three feature based on if it is Model 1 or 2, which is then processed by a fully connected (fc1) layer containing 16 neurons.

The number of neurons selected for the hidden layer was determined based on the configuration that yields the lowest test loss as shown in Fig. 4a. After evaluating models with hidden layer sizes ranging from 1 to 35 neurons, the optimal number of neurons was found to be 16, as it resulted in the lowest test loss. This indicates that a hidden layer with 16 neurons provides the best balance between model complexity and generalization performance, ensuring accurate predictions without

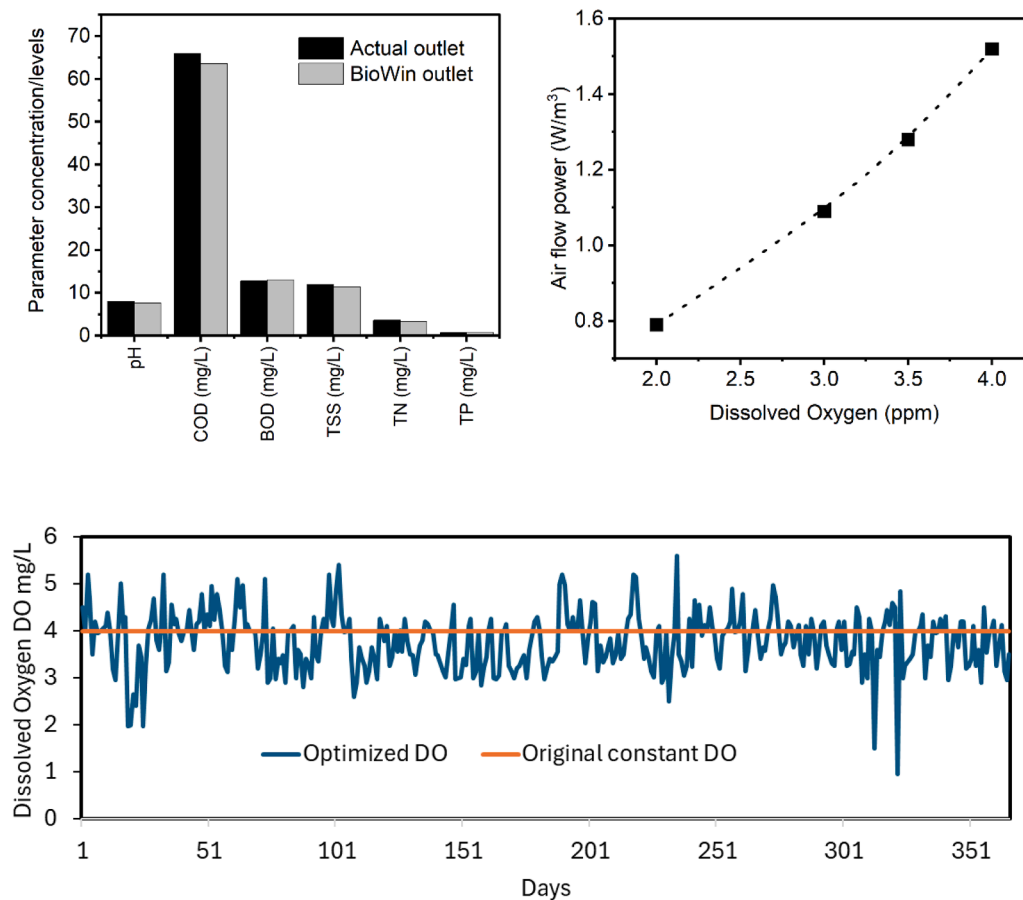


Fig. 5. a) BioWin model validation on effluent water quality parameters of STP. b) Airflow power with respect to DO c) Optimized DO versus original constant DO.

Table 3

Model performance evaluation: A comparison with some of the reported models for DO prediction.

Study Area	Model	Model evaluation metric	References
Danube River, Europe	Feed-forward back-propagation neural network	$R^2 \geq 0.86$	[32]
Main streams and chief tributaries, China	Multiple linear regression, artificial neural network and support vector machine	$R^2 \leq 0.76$	[33]
Lake Erie	Classification and regression tree	MAE = 1.262, RMSE = 2.117, NSE = 0.422	[34]
Canals, Bangkok	Artificial neural network	R = 0.84	[35]
River	Random Forest and LSTM	$R^2=0.744$, MAE = 0.528	[36]
–	Bayesian-Optimized Explainable Artificial Intelligence Approach	RMSE = 0.395, MAE = 0.303	[37]
Fanno Creek, Oregon	Deep recurrent neural network model	NSE= 0.948, RMSE = 0.43, MAE = 0.25	[38]
Fei-Tsui reservoir	Artificial neural network	$R^2= 0.98$	[39]
Fishery demonstration base in Jintan district, China	Combining the gradient boosting decision tree with the long short-term memory network	RMSE = 0.197, MAE = 0.299, MAPE= 0.092	[40]
Rock Creek waterway [Station no 01,648,010]	Hybrid machine learning models	MAPE=2.38, Pearson correlation coefficient = 0.9918	[41]
WWTP, Bahrain	Random Forest and Gradient Boost models	$R^2 = 1$, MAE = 0.01	[11]
Lake Taihu in Wuxi, China and a lake near Yixing, China	Deep learning and transfer learning	$R^2 = 0.793$, Index of agreement (d) = 0.89, NSE= 0.658	[42]
STP, South India	Model 1 (TSS+BOD+COD+Flowrate)	$R^2=0.8748$, Robust R-squared =0.9978, MAE = 0.1595	Present study
STP, South India	Model 2 (TSS+Flowrate)	$R^2 = 0.8738$, Robust R-squared = 0.9977, MAE = 0.159	Present study

overfitting.

The layer applies a Rectified Linear Unit (ReLU) activation function to introduce non-linearity, allowing the model to learn complex patterns in the data. A dropout layer with a 40 % dropout rate is included to prevent overfitting by randomly deactivating neurons during training. The output layer (fc2) consists of a single neuron, which produces the final prediction. The model is trained using the Adam optimizer with a learning rate of 0.001. The model undergoes training for 500 epochs to ensure convergence and optimal performance. The ANN model architecture is illustrated in the Fig. 4b

- g. Model Assessment: To evaluate the performance of the models, two of the metrics were used. The Mean Absolute Error (MAE) was chosen as the primary loss function during training, which measures the average absolute difference between the predicted and actual DO values. MAE was chosen as the loss function to minimize absolute deviations between predicted and actual values, ensuring robustness against outliers [29].

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - y_p| \quad (2.1)$$

n: Number of data points y_i : Actual value of the i th data point y_p : Predicted value of the i th data point

Additionally, the coefficient of determination (R^2) was used to assess the proportion of variance in the target variable (DO) that is predictable from the input features. An R^2 value closer to 1 indicates a better fit of the model to the data, showing that a significant portion of the variance in DO values can be explained by the model.

$$R^2 = \left(1 - \frac{\sum_{i=1}^n (y_i - y_p)^2}{\sum_{i=1}^n (y_i - y_{avg})^2} \right) \times 100\% \quad (2.2)$$

y_i : Actual value of the i th data point y_p : Predicted value of the i th data point y_{avg} : Mean of the actual values

During training, the model's loss was continuously monitored to track its convergence.

2.3.2. Process based: BioWin model

BioWin, a widely used software tool for modeling and simulating wastewater treatment processes was employed to digitally replicate the operations of the STP at Chikkabanavara.

Influent characteristics, such as wastewater flow rates, chamber dimensions, reaction rates and temperature (24 ± 1 °C) and pollutant concentrations, were plugged into BioWin to simulate the removal efficiencies of different treatment units [30]. Table 2 shows few important BioWin input parameters that were adjusted as per the field specific conditions. ANN Model predictions of DO were validated by comparing simulated effluent water quality parameters (e.g., BOD, COD, TSS, TN, TP) with measured values obtained from water quality monitoring on-field to ensure accuracy and reliability of ANN model.

In BioWin, oxygen transfer and microbial growth are modelled through a series of equations representing key biological and physical processes [31]. BioWin uses activated sludge models (like ASM1, ASM2, or ASM3) to simulate major of the microbial dynamics with Monod-type kinetics. Below are the key components related to oxygen transfer and microbial growth equations in BioWin:

a) Heterotrophic Growth: $r_H = \mu_H S_s \frac{X_H}{K_s + S_s}$ (Eq 2.3)

R_H denotes rate of heterotrophic growth, μ_H denotes maximum specific growth rate of heterotrophs. S_s denotes readily biodegradable substrate concentration, X_H denotes concentration of heterotrophic biomass and K_s denotes half-saturation constant for substrate.

b) Autotrophic Growth: $r_A = \mu_A S_{NH4} \frac{X_A}{K_A + S_{NH4}}$ (Eq 2.4)

Where r_A is Rate of autotrophic growth, μ_A is maximum specific growth rate of autotrophs. S_{NH4} and X_A represents concentrations of Ammonium and autotrophic biomass respectively. K_A is the half-saturation constant for ammonium.

c) Decay Rates: Biomass decay is represented as: $r_D = b \cdot X$ (Eq 2.5)

Where b is the decay coefficient, and X is the biomass concentration.

- d) Oxygen Demand: BioWin calculates oxygen demand based on Heterotrophic respiration (linked to COD/BOD removal) and Autotrophic respiration (linked to nitrification). The relationship between air flow rate (Q_{air}) and DO in an aeration chamber is governed by oxygen transfer dynamics, which are influenced by various factors such as the oxygen transfer efficiency (OTE), water depth, and operational conditions. A general expression is

$$DO_{out} = DO_{sat} - (DO_{sat} - DO_{in}) e^{(-k_{La} \cdot t)} \quad (2.6)$$

Where DO_{out} is Dissolved oxygen concentration leaving the chamber (mg/L) and DO_{in} denotes Dissolved oxygen concentration entering the chamber (mg/L). DO_{sat} is the saturation concentration of DO (mg/L) at given temperature and pressure, k_{La} represents Overall oxygen transfer coefficient (s^{-1}), which depends on the air flow rate (Q_{air}) and t signifies contact time (s). The oxygen transfer coefficient k_{La} is proportional to the air flow rate and can often be expressed as:

$$k_{La} = K \cdot Q_{air}^n \quad (2.7)$$

Where K is the proportionality constant dependent on system design and conditions, and n is the exponent typically ranging from 0.5 to 1, depending on diffuser type and mixing efficiency.

3. Results and discussion

3.1. BioWin model performance and insights on energy

First, BioWin was used to setup a digital twin of Chikkabanavara STP. Fig 5a shows the match between the experimentally obtained on-field data and the BioWin predicted data, was within 5 % error limits. Thus, the digital twin model developed could capture accurately the complex biological processes in Chikkabanavara STP. This model was later used to assess the performance of ANN models.

BioWin was also used to translate the DO optimization into corresponding energy savings. From an energy perspective, DO maintenance at 2 ppm requires approximately 50 % less energy compared to 4 ppm (as seen from Fig 5b). Given that aeration processes typically account for a substantial portion of energy costs in wastewater treatment plants, this reduction offers a pathway to substantial operational savings.

3.1.1. Quantitative analysis of energy savings

The DO optimization in Chikkabanavara STP demonstrated considerable potential for reducing aeration energy consumption. By dynamically adjusting DO levels to match varying influent characteristics, aeration energy usage was reduced by an estimated 9.4 %, leading to significant cost savings. This improvement stems from maintaining an average yearly DO level of 3.7 mg/L instead of the standard 4 mg/L, resulting in an energy savings of 0.15 W/m³ (as shown in Fig 5c). Furthermore, the study quantifies energy savings associated with optimized aeration: a 3.63 MLD STP could save 0.15 W/m³, translating to approximately ₹1 million annually at an average electricity tariff of ₹7.11/kWh. These findings highlight both the scientific innovation and real-world economic impact of our data-driven strategy for enhancing energy efficiency in wastewater treatment.

3.2. ANN model performance: benchmarking with state of art and implications for practice

Table 3 shows that various models have been employed for DO prediction in diverse aquatic matrices with diverse performance metrics. Traditional models like multiple linear regression and support vector machines exhibited lower predictive accuracy ($R^2 \leq 0.76$). Several AI ML models have also showed $0.74 < R^2 < 1$. To the best of authors knowledge only a few studies in literature have reported optimization of DO for STP [43–45].

The comparison between Model 1 (TSS + BOD + COD + Flowrate) and Model 2 (TSS only + Flowrate) reveals that both models exhibit similar predictive performance, with only marginal differences in error metrics. The MAE is nearly identical for both models (0.1595 vs. 0.1597), indicating that the average deviation of predictions from actual values remains consistent. Likewise, the R^2 values 0.8748 for Model 1 and 0.8738 for Model 2 suggest that both models explain approximately the same proportion of variance in the dataset. The Robust R-squared (R^2_s) values (0.9978 vs. 0.9977) suggest that both models perform exceptionally well under robust regression analysis, reinforcing their reliability in predicting organic load. Fig 6 shows the comparison between the on-field estimated DO values and model predicted DO values. The slight difference in performance indicates that while adding BOD and COD slightly improves model accuracy, TSS alone remains a viable and efficient proxy for COD, BOD.

3.2.1. Implications for practice

COD and BOD are crucial parameters for assessing organic load in wastewater. However, their measurement is often labour-intensive, time-consuming, and requires specialized equipment. TSS, which measures the concentration of suspended particles in wastewater, often correlates with organic matter content, as a significant portion of these solids is biodegradable. By establishing site-specific correlations between TSS and COD/BOD, wastewater treatment plants can use TSS as an indirect indicator of organic load. This approach allows for more frequent monitoring and rapid decision-making, optimizing treatment processes while reducing the need for extensive COD/BOD testing. Using TSS as a proxy for COD/BOD not only simplifies monitoring but also aids in process control, ensuring efficient operation and compliance with discharge standards allowing for more frequent monitoring and faster decision-making in wastewater treatment without the need for labour-intensive COD/BOD measurements. These findings support the practical use of TSS as an indirect indicator of COD/BOD, especially in municipal sewage. This approach of using TSS as proxy for COD/BOD is unsuitable for effluent treatment plants where inorganic load is significantly higher.

3.3. Validating the proposed technological solution: A BioWin approach

In Section 3.2, the ANN model was used to predict optimized DO with respect to water flow rates and water quality. But, to ensure that the predicted DO does in fact lead to efficient wastewater treatment, the validated BioWin model (Section 3.1) was used to predict scenario-based effluent quality. Table 4 compares BioWin-predicted outlet [with ANN model generated DO] and real outlet. These cases were selected for validation to encompass a range of flow rates and influent quality conditions. The high Robust R^2 value of 0.99 indicates that the model outperforms several existing models in predicting DO for mid-range values. Additionally, the $R^2 = 0.87$ results demonstrate that the model accurately captures DO levels even under extreme conditions. The BioWin results demonstrate a strong alignment with the actual effluent quality observed in the field, thereby verifying the model's accuracy.

By integrating ANN predictions with BioWin simulations, the comparison highlights the ANN model's ability to generate rapid predictions while reinforcing the necessity of mechanistic validation to ensure

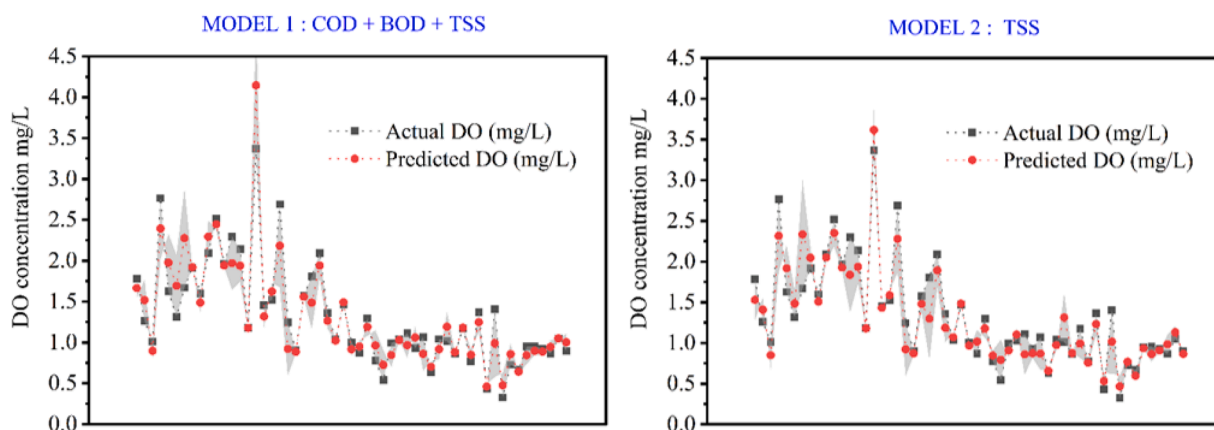


Fig. 6. a) Actual DO vs model 1 predicted DO b) Actual DO vs model 2 predicted DO.

Table 4

Case study: comparison of BioWin- predicted outlet [with ANN model generated DO] and real outlet.

Q = 4.03 MLD					
Parameters	Inlet water quality	BioWin predicted outlet water quality based on Model 1 output DO (4.14 mg/L).	BioWin predicted outlet water quality based on Model 2 output DO (3.61 mg/L).	Actual outlet water quality Actual DO (3.37 mg/L)	Discharge standards to inland surface water [46]
BOD (mg/L)	290	13.5	13.55	13.8	30
COD (mg/L)	680	57	57.6	58.8	250
NH4-N (mg/L)	26	0.5	0.5	0.6	50
TP (mg/L)	3	1	0.9	0.9	5
pH	7.11	7.6	7.6	7.7	5.5 –9
TSS (mg/L)	241	12.5	12.7	13	100
Q = 2.33 MLD					
Parameters	Inlet water quality	BioWin predicted outlet water quality based on Model 1 output DO (1.06 mg/L).	BioWin predicted outlet water quality based on Model 2 output DO (0.87 mg/L).	Actual outlet water quality Actual DO (0.93 mg/L)	Discharge standards to inland surface water [46]
BOD (mg/L)	296	10.2	10.2	10.3	30
COD (mg/L)	580	67	67.5	68	250
NH4-N (mg/L)	27.1	0.8	0.8	0.8	50
TP (mg/L)	4.1	0.6	0.64	0.7	5
pH	7.08	7.7	7.6	7.7	5.5 –9
TSS (mg/L)	200	10.7	10.8	10.8	100
Q = 3.68 MLD					
Parameters	Inlet water quality	BioWin predicted outlet water quality based on Model 1 output DO (1.06 mg/L).	BioWin predicted outlet water quality based on Model 2 output DO (0.87 mg/L).	Actual outlet water quality Actual DO (0.93 mg/L)	Discharge standards to inland surface water [46]
BOD (mg/L)	320	11.0	11.0	10.8	30
COD (mg/L)	690	61	61.1	60.8	250
NH4-N (mg/L)	24.6	0.7	0.7	0.9	50
TP (mg/L)	2.9	0.59	0.6	0.6	5
pH	6.81	7.6	7.6	7.67	5.5 –9
TSS (mg/L)	211	13.6	13.8	13	100
BOD (mg/L)	320	11.0	11.0	10.8	30

practical applicability in STP operations. The hybrid framework enhances confidence in data-driven modeling while ensuring compliance with process-based wastewater treatment principles.

4. Conclusions and future perspectives

Dynamic dissolved oxygen management based on real-time inflow fluctuations reduces operational costs, improves treatment performance, and contributes to achieving Sustainable Development Goals (SDGs) 6 and 7.

This study highlights that TSS can serve as a practical proxy for water quality parameters like COD and BOD, for training ANN model to predict optimized DO for treating of municipal sewage. The ANN model achieves comparable accuracy when using TSS as the as a standalone indicator of water quality along with water flow rate, making it a cost-effective and simpler/easy-to-measure alternative for real time water quality monitoring and process optimization.

In the future, alongside the prediction of optimized DO, the prediction of optimized Mixed Liquor Suspended Solids (MLSS) can be explored using ANNs. This approach holds the potential to significantly enhance the efficiency and speed of water treatment processes.

CRediT authorship contribution statement

Reshmi Das: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Vishnudatha Venu:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation,

Conceptualization. **Lakshminarayana Rao:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.cep.2025.110599](https://doi.org/10.1016/j.cep.2025.110599).

Data availability

Data will be made available on request.

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